

Digital Signal Analysis and Processing Numerical Companion

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

Ch 2 · Z-Transform

Numerical Problems & Solutions

275 numerical marks, 7.5% of the archive, 82% of this chapter · **32 of the 45 papers ask for an inverse Z-transform.** Every one of them is below. They split into five methods; the method is worked once in full and the other papers follow with their own data and answer.

The one procedure, used in §1, §2, §3 and §5

- 1. Proper?** If the numerator's degree is $\geq$ the denominator's, divide first. The quotient is a polynomial and becomes impulses; the remainder is proper.
- 2. Divide by z ,** expand $X(z)/z$ in partial fractions, multiply back by z . Every term then reads $\frac{Az}{z-a}$, which is a standard pair.
- 3. Assign each pole a side** from the ROC: pole radius *below* the ROC $\rightarrow$ right-sided; *above* $\rightarrow$ left-sided.
- 4. Write the answer term by term.**

The pairs (the whole table)

Term	$ z > a $	$ z < a $
$\frac{z}{z-a}$	$a^n u[n]$	$-a^n u[-n-1]$
$\frac{z}{(z-a)^2}$	$n a^{n-1} u[n]$	$-n a^{n-1} u[-n-1]$
C	$C \delta[n]$	
z^k	$\delta[n+k]$	
z^{-k}	$\delta[n-k]$	

The three checks that cost nothing

- 1. Initial value:** for a causal $x[n]$, $x[0] = \lim_{z \rightarrow \infty} X(z)$.
- 2. Sum of residues:** the partial-fraction constants must add back to the original numerator's leading coefficient.
- 3. Substitute one number:** compute $x[0]$ or $x[1]$ from your answer and from a direct long division of $X(z)$. If they disagree, the ROC assignment is the usual culprit.

A right-sided answer that contains $u[-n-1]$, or a causal answer with samples at negative n , is wrong on sight.

1 Partial fractions, and how the ROC picks the answer

TOP 10/45 82 Ba **1+5**, 81 Ba **2+2+3**, 73 Bh **6**, 72 Ma **6**, 71 Ch **1+5**, 76 Bh **2+6**, 77 Ch **6**, 70 Ma **6**, 80 Ch **6**, 70 Bh **6** · **one $X(z)$, three answers:** the point of the whole chapter

QUESTION AS PRINTED 2082 Baishakh · Q3 · [1+5]

Define the ROC. Find the inverse z-transform of $X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$ using partial fraction method for (a) $|z| > 1$ (b) $|z| < 0.5$ (c) $0.5 < |z| < 1$.

Step 1: factor the denominator and mark the poles

- $1 - 1.5z^{-1} + 0.5z^{-2} = (1 - z^{-1})(1 - 0.5z^{-1})$, so the poles are at $z = 1$ and $z = 0.5$.
- Numerator degree 0 < denominator degree 2, so it is already proper: no division.
- Three ROCs are possible, and the question asks for all three: $|z| < 0.5$, $0.5 < |z| < 1$, $|z| > 1$.

Step 2: partial fractions, once, for all three parts

$$\frac{1}{(1 - z^{-1})(1 - 0.5z^{-1})} = \frac{A}{1 - z^{-1}} + \frac{B}{1 - 0.5z^{-1}}$$

- Put $w = z^{-1}$: $1 = A(1 - 0.5w) + B(1 - w)$.
- $w = 1$ kills B : $1 = A(1 - 0.5) \Rightarrow \boxed{A = 2}$.
- $w = 2$ kills A : $1 = B(1 - 2) \Rightarrow \boxed{B = -1}$.
- Check: $A + B = 1$, the numerator's constant term ✓

$$X(z) = \frac{2}{1 - z^{-1}} - \frac{1}{1 - 0.5z^{-1}}$$

Step 3: the ROC decides each term's side, and nothing else changes

Part	pole $z = 1$	pole $z = 0.5$	$x[n]$
(a) $ z > 1$	outside both $\rightarrow$ right	right	$[2 - (0.5)^n]u[n]$
(b) $ z < 0.5$	inside both $\rightarrow$ left	left	$[-2 + (0.5)^n]u[-n - 1]$
(c) $0.5 < z < 1$	ROC is inside it $\rightarrow$ left	ROC is outside it $\rightarrow$ right	$-2u[-n - 1] - (0.5)^n u[n]$

Reading the three answers

- (a) is **causal**: $x[0] = 1, x[1] = 1.5, x[2] = 1.75 \dots$ climbing to 2. Contains the unit circle? **No**: so this one is not stable.
- (b) is **anticausal**: $x[-1] = 0, x[-2] = 2, x[-3] = 6$, growing as $n \rightarrow -\infty$.
- (c) is **two-sided** and is the only one whose ROC contains $|z| = 1$, so **(c) is the stable one**: $x[0] = -1, x[1] = -0.5$, and $x[n] = -2$ for every $n \leq -1$.
- **All three come from the same two fractions**. Only the choice of pair changes. That is the whole examinable idea.
- If the question gives *one* ROC, do steps 1 and 2 identically and read one row of the table.
- If it says "**causal**" instead of giving an ROC, that means the outermost row: every pole right-sided.

Every other paper that runs this same method

Paper	Question as printed	$x[n]$
81 Ba 2+2+3	Determine the inverse z-transform of $X(z) = \frac{1}{1 - 0.8z^{-1} + 0.12z^{-2}}$. (i) if ROC is $ z > 0.6$ (ii) if ROC is $ z < 0.2$ (iii) if ROC is $0.2 < z < 0.6$ poles 0.6 and 0.2; $A = 1.5, B = -0.5$	(i) $[1.5(0.6)^n - 0.5(0.2)^n]u[n]$ (ii) $[-1.5(0.6)^n + 0.5(0.2)^n]u[-n - 1]$ (iii) $-1.5(0.6)^n u[-n - 1] - 0.5(0.2)^n u[n]$
73 Bh 6 , 72 Ma 6	73 Bh: Using long division determine the inverse Z-transform of, $X(z) = \frac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$ when, ROC: $ Z > 1$ and ROC: $ Z < \frac{1}{2}$ 72 Ma: Determine the inverse Z-transform of, $X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$ When, (a) ROC: $ Z > 1.5$ (b) ROC: $ Z < 0.5$ $\frac{3}{2}$ and $\frac{1}{2}$ are 1.5 and 0.5, so both are the worked $X(z)$	$[2 - (0.5)^n]u[n]$ $[-2 + (0.5)^n]u[-n - 1]$
71 Ch 1+5	Define ROC. Find inverse Z-transform of $X(z) = \frac{1}{(z - 0.5)(z + 2)}$, if i) ROC: $0.5 < z < 2$ ii) ROC: $ z < 0.5$ iii) ROC: $ z > 2$ divide by z first, giving $-1 + \frac{0.8z}{z - 0.5} + \frac{0.2z}{z + 2}$	(i) $-\delta[n] + 0.8(0.5)^n u[n] - 0.2(-2)^n u[-n - 1]$ (ii) $-\delta[n] - [0.8(0.5)^n + 0.2(-2)^n]u[-n - 1]$ (iii) $-\delta[n] + [0.8(0.5)^n + 0.2(-2)^n]u[n]$

71 Ch is the one that needs the $X(z)/z$ step: its numerator has no z , so expanding directly would leave $\frac{A}{z-a}$ terms. Dividing by z first produces the constant -1 (an impulse) plus two proper $\frac{Az}{z-a}$ terms. 72 Ma prints its first ROC as $|z| > 1.5$; for this $X(z)$ that is the same region as $|z| > 1$, since both lie outside every pole, so the answer is unchanged.

2 Improper in z^{-1} : divide first, the quotient becomes impulses

TOP 7/45 80 Bh **1+6**, 80 Ba **1+5**, 71 Shr **6**, 69 Bh **6**, 80 Ch **6**, 70 Bh **6**, 70 Ma **6** · **numerator and denominator both degree 2 in z^{-1}**

QUESTION AS PRINTED 2080 Bhadra · Q3 · [1+6]

Define ROC. Find the inverse z-transform of $H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}}$, ROC: $0.25 < |z| < 0.5$.

Step 1: spot that it is improper, and divide

- Both numerator and denominator have degree 2 in z^{-1} , so a constant must come out first. Divide the *highest* powers: $z^{-2} \div 0.125z^{-2} = 8$.

$$N - 8D = (1 - 8) + (2 + 6)z^{-1} + (1 - 1)z^{-2} = -7 + 8z^{-1}$$

$$H(z) = 8 + \frac{-7 + 8z^{-1}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

- **The 8 is already an answer:** a constant is $8\delta[n]$.

Step 2: factor, then partial fractions on the remainder

- $1 - 0.75z^{-1} + 0.125z^{-2} = (1 - 0.5z^{-1})(1 - 0.25z^{-1})$: poles 0.5 and 0.25.
- With $w = z^{-1}$: $-7 + 8w = A(1 - 0.25w) + B(1 - 0.5w)$.
- $w = 2$: $-7 + 16 = 9 = A(0.5) \Rightarrow A = 18$. $w = 4$: $-7 + 32 = 25 = B(-1) \Rightarrow B = -25$.
- Check $A + B = -7 \checkmark$

$$H(z) = 8 + \frac{18}{1 - 0.5z^{-1}} - \frac{25}{1 - 0.25z^{-1}}$$

Step 3: the ROC $0.25 < |z| < 0.5$ splits the two poles

- Pole 0.5: the ROC lies *inside* it $\rightarrow$ **left-sided** $\rightarrow -18(0.5)^n u[-n-1]$.
- Pole 0.25: the ROC lies *outside* it $\rightarrow$ **right-sided** $\rightarrow -25(0.25)^n u[n]$.
- **Check:** $h[0] = 8 + 0 - 25 = -17$;
 $h[1] = -25(0.25) = -6.25$; $h[-1] = -18(2) = -36$;
 $h[-2] = -18(4) = -72$.
- The ROC is a ring not containing $|z| = 1$, so this system is **neither causal nor stable**: worth one line if the question asks.

$$h[n] = 8\delta[n] - 18(0.5)^n u[-n-1] - 25(0.25)^n u[n]$$

Every other paper that runs this same method

Paper	Question as printed	After dividing	$x[n]$
80 Ba 1+5	Define ROC. Find inverse z-transform of $X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 1.5z^{-1} + 0.5z^{-2}}$, ROC: $ z > 1$.	$2 + \frac{8}{1 - z^{-1}} - \frac{9}{1 - 0.5z^{-1}}$	$2\delta[n] + [8 - 9(0.5)^n]u[n]$
71 Shr 6	Find the inverse z transform of $X(Z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + 1.5z^{-1} + 0.5z^{-2}}$, $ z > 1$ using partial fraction method. note the +1.5: it is not 80 Ba's denominator	$2 - \frac{1}{1 + 0.5z^{-1}}$ one factor cancels	$2\delta[n] - (-0.5)^n u[n]$
69 Bh 6	An LTI system is characterized by the system function: $H(z) = \frac{1 - \frac{1}{2}Z^{-2}}{(1 - \frac{1}{2}Z^{-1})(1 - \frac{1}{4}Z^{-1})}$, $ Z > \frac{1}{2}$. De-terminate the impulse response of the system. "impulse response" is the inverse transform under another name	$-4 + \frac{-2}{1 - 0.5z^{-1}} + \frac{7}{1 - 0.25z^{-1}}$	$-4\delta[n] + [-2(0.5)^n + 7(0.25)^n]u[n]$
80 Ch, 70 Bh 6	Determine the causal signal $x[n]$ if its z-transform $X(z)$ is given by $X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + 4z^{-1} + 4z^{-2}}$ "causal" is the ROC: it means $ z > 2$, and no ROC is printed	$0.25 + \frac{0.5}{1 + 2z^{-1}} + \frac{0.25}{(1 + 2z^{-1})^2}$	$0.25\delta[n] + (0.25n + 0.75)(-2)^n u[n]$

71 Shr is the free one. Its numerator is $(1 + z^{-1})^2$ and its denominator $(1 + z^{-1})(1 + 0.5z^{-1})$, so after the division the remainder $-1 - z^{-1}$ is exactly $-(1 + z^{-1})$ and **one factor cancels**. Look for a common factor before grinding through partial fractions.

80 Ch / 70 Bh has a repeated pole. Its denominator is $(1 + 2z^{-1})^2$, a double pole at $z = -2$. Match $0.75 + w = A(1 + 2w) + B$ to get $A = 0.5$, $B = 0.25$, then use $\frac{1}{(1 - az^{-1})^2} \leftrightarrow (n+1)a^n u[n]$. "Causal" in the question replaces a stated ROC and means $|z| > 2$.

3 Improper in z : the z^4/z^2 family

TOP 11/45 79 Bh 6, 76 Ch 2+6, 81 Ch 6, 71 Bh 6, 79 Ch 6, 72 Ash 5, 81 Bh 1+5, 74 Ch 1+3+5, 73 Ch 1+3+5, 74 Bh 1+3+5 · the same question with the numerator changed, eleven times

QUESTION AS PRINTED 2079 Bhadra · Q3 · [6]

Find the inverse z -transform of $X(z) = \frac{2z^4 + 2z^3 - 3z + 2}{z^2 - 1.5z - 1}$, ROC: $|z| < 0.5$, using partial fraction expansion.

Step 1: poles, and what the ROC means

- $z^2 - 1.5z - 1 = (z - 2)(z + 0.5)$: poles at $z = 2$ and $z = -0.5$, radii 2 and 0.5.
- ROC $|z| < 0.5$ is inside *both*, so **both poles are left-sided** and the whole answer carries $u[-n - 1]$.

Step 2: long division in z , because the numerator is degree 4

Written as ordinary polynomial division, highest power first. The missing z^2 term is carried as $0z^2$.

Quotient term	subtract	new remainder
$2z^2$	$2z^2(z^2 - 1.5z - 1) = 2z^4 - 3z^3 - 2z^2$	$5z^3 + 2z^2 - 3z + 2$
$5z$	$5z(z^2 - 1.5z - 1) = 5z^3 - 7.5z^2 - 5z$	$9.5z^2 + 2z + 2$
9.5	$9.5(z^2 - 1.5z - 1) = 9.5z^2 - 14.25z - 9.5$	$16.25z + 11.5$

$$X(z) = \underbrace{2z^2 + 5z + 9.5}_{\text{quotient}} + \underbrace{\frac{16.25z + 11.5}{(z - 2)(z + 0.5)}}_{G(z), \text{ proper}}$$

Step 3: divide $G(z)$ by z and expand

$$\frac{G(z)}{z} = \frac{16.25z + 11.5}{z(z - 2)(z + 0.5)} = \frac{C}{z} + \frac{D}{z - 2} + \frac{E}{z + 0.5}$$

- $C = \frac{11.5}{(-2)(0.5)} = -11.5$
- $D = \frac{16.25(2) + 11.5}{2(2.5)} = \frac{44}{5} = 8.8$
- $E = \frac{16.25(-0.5) + 11.5}{(-0.5)(-2.5)} = \frac{3.375}{1.25} = 2.7$

$$G(z) = -11.5 + \frac{8.8z}{z - 2} + \frac{2.7z}{z + 0.5}$$

Step 4: apply the pairs

- $2z^2 \rightarrow 2\delta[n + 2]$, $5z \rightarrow 5\delta[n + 1]$, $9.5 \rightarrow 9.5\delta[n]$ (**powers of z are advances**).
- $-11.5 \rightarrow -11.5\delta[n]$, which merges with the 9.5 to give $-2\delta[n]$.
- Both poles left-sided, so $\frac{Az}{z - a} \rightarrow -A a^n u[-n - 1]$.

Check, sample by sample

n	from the answer	$x[n]$
1	nothing is non-zero	0
0	-2	-2
-1	$5 - [8.8(0.5) + 2.7(-2)]$	6
-2	$2 - [8.8(0.25) + 2.7(4)]$	-11
-3	$0 - [8.8(0.125) + 2.7(-8)]$	20.5

$x[n] = 0$ for all $n \geq 1$, as a purely left-sided answer must be. If your answer has a tail to the right, the ROC was read backwards.

Every other paper in this family

Identical denominator $(z - 2)(z + 0.5)$ and identical ROC $|z| < 0.5$; only the numerator changes. Q is the quotient, and the impulse at $n = 0$ is $Q_0 + C$ after C is folded in.

Paper	Question as printed	Quotient $Q(z)$, and C	$x[n]$
76 Ch 2+6	Define ROC of z-transform. Find inverse z-transform using partial fraction expansion of $X(z) = \frac{z^4 + 5z^3 - 3z + 4}{z^2 - 1.5z - 1}$, ROC: $ z < 0.5$.	$z^2 + 6.5z + 10.75$; $C = -14.75$	$\delta[n+2] + 6.5\delta[n+1] - 4\delta[n] - [10.8(2)^n + 3.95(-0.5)^n]u[-n-1]$
81 Ch, 71 Bh 6	Determine the inverse Z-transform of $X(z) = \frac{z^4 + z^3 - 3z + 5}{z^2 - 1.5z - 1}$ When ROC: $ z < 0.5$ 71 Bh prints the identical question	$z^2 + 2.5z + 4.75$; $C = -9.75$	$\delta[n+2] + 2.5\delta[n+1] - 5\delta[n] - [4.6(2)^n + 5.15(-0.5)^n]u[-n-1]$
79 Ch 6	Find the inverse Z-transform of $X(z) = \frac{z^4 - 2z^3 - z + 4}{z^2 - 1.5z - 1}$ ROC: $ z < 0.5$	$z^2 - 0.5z + 0.25$; $C = -4.25$	$\delta[n+2] - 0.5\delta[n+1] - 4\delta[n] - [0.4(2)^n + 3.85(-0.5)^n]u[-n-1]$
72 Ash 5	Find inverse Z-transform of $X(z) = \frac{z^4 + 2z^3 - z + 4}{z^2 - 1.5z - 1}$, ROC: $ z < 0.5$	$z^2 + 3.5z + 6.25$; $C = -10.25$	$\delta[n+2] + 3.5\delta[n+1] - 4\delta[n] - [6.8(2)^n + 3.45(-0.5)^n]u[-n-1]$

74 Ch, 73 Ch, 74 Bh 1+3+5 : different denominator

- $X(z) = \frac{2z^3 + 2z^2 + 3z + 5}{z^2 - 0.1z - 0.2}$, ROC $|z| < 0.4$.
- All three papers misprint the leading term as $2z^2$, which would make the fraction proper and the question trivial. Solved here as $2z^3$. verified/added
- $z^2 - 0.1z - 0.2 = (z - 0.5)(z + 0.4)$, both left-sided under $|z| < 0.4$.
- Division gives $Q = 2z + 2.2$, remainder $\frac{3.62z + 5.44}{(z - 0.5)(z + 0.4)}$.
- $C = -27.2$, $D = 16.111$, $E = 11.089$.

$$x[n] = 2\delta[n+1] - 25\delta[n] - [16.111(0.5)^n + 11.089(-0.4)^n]u[-n-1]$$

- Check: $x[0] = -25$, $x[-1] = -2.5$, $x[-2] = -133.75$.

81 Bh 1+5 : the clean one

- $X(z) = \frac{2z^3 - 5z^2 + z + 3}{(z - 1)(z - 2)}$, ROC $|z| < 1$: both poles left-sided.
- Denominator = $z^2 - 3z + 2$. Division gives quotient $2z + 1$ and **remainder exactly 1**:

$$X(z) = 2z + 1 + \frac{1}{(z - 1)(z - 2)}$$

$$- \frac{1}{z(z - 1)(z - 2)} = \frac{0.5}{z} - \frac{1}{z - 1} + \frac{0.5}{z - 2}, \text{ so}$$

$$\frac{1}{(z - 1)(z - 2)} = 0.5 - \frac{z}{z - 1} + \frac{0.5z}{z - 2}$$

$$x[n] = 2\delta[n+1] + 1.5\delta[n] + [1 - 0.5(2)^n]u[-n-1]$$

- Check: $x[0] = 1.5$, $x[-1] = 2.75$, $x[-2] = 0.875$, $x[-3] = 0.9375$.

4 Inverse by long division (power series)

HOT 5/45 73 Bh 6, 72 Ma 6, 76 Bh 2+6, 77 Ch 6, 70 Ma 6 · only when the question names the method

QUESTION AS PRINTED 2073 Bhadra · Q3 · [6]

Determine the inverse z-transform of $X(z) = \frac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$ using long division method, when ROC: $|z| > 1$ and when ROC: $|z| < \frac{1}{2}$.

Method (the direction of the division is the whole question)

1. ROC outside → right-sided → arrange both polynomials in **ascending powers of z^{-1}** and divide.
2. ROC inside → left-sided → rewrite in **ascending powers of z** first, then divide.
3. Read the quotient's coefficients off as $x[0], x[1], x[2] \dots$ (or $x[-1], x[-2] \dots$).
4. **Give the recurrence too**: a division gives numbers, and the recurrence is what shows the pattern continues.

(a) ROC $|z| > 1$, right-sided

- Divide 1 by $1 - 1.5z^{-1} + 0.5z^{-2}$ in ascending z^{-1} :

step	quotient term	remainder
1	1	$1.5z^{-1} - 0.5z^{-2}$
2	$1.5z^{-1}$	$1.75z^{-2} - 0.75z^{-3}$
3	$1.75z^{-2}$	$1.875z^{-3} - 0.875z^{-4}$
4	$1.875z^{-3}$	...

$$x[n] = \{1, 1.5, 1.75, 1.875, 1.9375, \dots\}$$

- Recurrence from the denominator:
 $x[n] = 1.5x[n-1] - 0.5x[n-2]$, $x[0] = 1$.
- **Closed form, if you want it:** partial fractions give
 $x[n] = [2 - (0.5)^n]u[n]$, which matches every term. This is 82 Ba part (a).

(b) ROC $|z| < \frac{1}{2}$, left-sided

- Rewrite with z ascending: multiply top and bottom by z^2 ,

$$X(z) = \frac{z^2}{z^2 - 1.5z + 0.5} = \frac{z^2}{0.5 - 1.5z + z^2}$$

and divide z^2 by $0.5 - 1.5z + z^2$.

- First term $\frac{z^2}{0.5} = 2z^2$, and the division continues in rising powers of z .

$$x[n] = \{\dots, 6, 2, 0, 0\} \quad \text{for } n = -3, -2, -1, 0$$

- **Closed form:** $x[n] = [-2 + (0.5)^n]u[-n-1]$, so
 $x[-1] = 0$, $x[-2] = 2$, $x[-3] = 6$, $x[-4] = 14$.
- $x[-1] = 0$ is not a mistake: the two terms cancel exactly at $n = -1$.

The other three, and the two ROCs that cannot be right

Paper	Question as printed	Answer
72 Ma 6	Determine the inverse Z-transform of, $X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$ When, (a) ROC: $ z > 1.5$ (b) ROC: $ z < 0.5$	Same as 73 Bh: $ z > 1.5$ is outside every pole, so it is the same region as $ z > 1$
76 Bh 2+6	List out the properties of z-transform. Using long division, determine the inverse Z-transform of, $X(Z) = \frac{1}{1 - 1.58z^{-1} + 0.5z^{-2}}$ when ROC: $ z > 1$ and ROC: $ z < \frac{1}{2}$.	Right-sided division gives $\{1, 1.58, 1.9964, 2.3643, 2.7374, \dots\}$ with $x[n] = 1.58x[n-1] - 0.5x[n-2]$
77 Ch 6	Find the inverse z-transform for $(z) = \frac{1}{1 - 0.5z^{-1} + 1.5z^{-2}}$, ROC: $ z > 1$ using long division method. the paper prints " (z) " with no X	$\{1, 0.5, -1.25, -1.375, 1.1875, 2.65625\}$, $x[n] = 0.5x[n-1] - 1.5x[n-2]$
70 Ma 6	Determine the inverse z-transform of $H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + 4z^{-1} + 4z^{-2}}$ Use long division method for possible ROCs.	$ z > 2$: the causal answer of §2. $ z < 2$: the same partial fractions with $u[-n-1]$ and a sign change

76 Bh: the printed ROCs are impossible

- 1.58 puts the poles at $z = 1.1423$ and $z = 0.4377$, not at 1 and 0.5.
- So " $|z| > 1$ " contains the pole at 1.1423, and " $|z| < \frac{1}{2}$ " contains the pole at 0.4377. **An ROC can never contain a pole** (§2.2, property 3), so neither region is a legal ROC.
- With 1.5 instead of 1.58 the poles are exactly 1 and 0.5 and both printed ROCs are correct: and 73 Bh asks the identical question with $\frac{3}{2}$. **1.58 is a misprint for 1.5.** verified/added
- In the exam: divide as printed (the division does not care), and add one line saying the ROC must be $|z| > 1.1423$ or $|z| < 0.4377$.

77 Ch: same defect, different cause

- $z^2 - 0.5z + 1.5$ has complex roots, $z = 0.25 \pm j1.198$, both of modulus $\sqrt{1.5} = 1.2247$.
- "ROC $|z| > 1$ " again contains them, so it is not a legal ROC; the causal ROC is $|z| > 1.2247$. verified/added
- The poles are outside the unit circle, so this sequence **grows**: the printed terms alternate and increase in size, which is correct, not an arithmetic slip.
- **Long division is unaffected** by all of this: it only needs to know the sequence is right-sided.

5 Repeated poles: the $X(z)/z$ method earning its keep

HOT 3/45 76 Ash 3+6, 73 Shr 1+5, 70 Asa 1+5

QUESTION AS PRINTED 2076 Ashwin · Q3 · [3+6]

State Convolution property of Z-transform. Find inverse Z-transform of $X(z) = \frac{z}{(z - 0.6)(z + 0.5)^2}$, ROC: $|z| > 0.6$.

Step 1: divide by z , which removes the numerator entirely

$$\frac{X(z)}{z} = \frac{1}{(z - 0.6)(z + 0.5)^2} = \frac{A}{z - 0.6} + \frac{B}{z + 0.5} + \frac{C}{(z + 0.5)^2}$$

- A : cover $(z - 0.6)$ and put $z = 0.6$: $A = \frac{1}{(0.6 + 0.5)^2} = \frac{1}{1.21} = 0.8264$.
- C : cover $(z + 0.5)^2$ and put $z = -0.5$: $C = \frac{1}{-0.5 - 0.6} = \frac{1}{-1.1} = -0.9091$.
- B : clear denominators, $1 = A(z + 0.5)^2 + B(z - 0.6)(z + 0.5) + C(z - 0.6)$, and match the z^2 coefficient: $0 = A + B$, so $B = -A = -0.8264$.
- Matching one coefficient is faster than differentiating, and it is the step to use for every repeated pole.

Step 2: multiply back by z and apply the pairs

$$X(z) = \frac{0.8264z}{z - 0.6} - \frac{0.8264z}{z + 0.5} - \frac{0.9091z}{(z + 0.5)^2}$$

- ROC $|z| > 0.6$ is outside both poles, so **every term is right-sided**.
- $\frac{z}{z - a} \rightarrow a^n u[n]$ and $\frac{z}{(z - a)^2} \rightarrow n a^{n-1} u[n]$, here with $a = -0.5$.
- Tidier, using $(-0.5)^{n-1} = -2(-0.5)^n$:

$$x[n] = \left[0.8264(0.6)^n - 0.8264(-0.5)^n - 0.9091 n(-0.5)^{n-1} \right] u[n]$$

Check

n	substitution	$x[n]$
0	$0.8264 - 0.8264$	0
1	$0.4959 + (1.8182 - 0.8264)(-0.5)$	0
2	$0.2975 + (3.6364 - 0.8264)(0.25)$	1
3	$0.1785 + (5.4545 - 0.8264)(-0.125)$	-0.4

$x[0] = x[1] = 0$ is correct, not a slip. $X(z)$ has one z on top and a cubic underneath, so the first non-zero sample cannot appear before $n = 2$: a quick sanity check before any arithmetic.

The two left-sided siblings

Paper	Question as printed	Constants	$x[n]$
73 Shr 1+5	Define a ROC. Find inverse Z-transform of $X(z) = \frac{z}{(z - 0.4)(z + 1.5)^2}$, ROC: $ z < 0.4$	$A = \frac{1}{3.61} = 0.2770$, $B = -0.2770$, $C = \frac{1}{-1.9} = -0.5263$	$\left[-0.2770(0.4)^n + 0.2770(-1.5)^n + 0.5263 n(-1.5)^{n-1} \right] u[-n - 1]$
70 Asa 1+5	Define Region of Convergence. Find inverse Z-transform of $X(z) = \frac{z}{(z - 1)(z - 2)^2}$, ROC: $ z < 1$	$A = 1, B = -1, C = 1$	$\left[-1 + (2)^n - n(2)^{n-1} \right] u[-n - 1]$

Both ROCs lie inside every pole, so every term flips to $u[-n - 1]$ with a sign change: including the repeated one. Check 70 Asa: $x[-1] = -1 + 0.5 + 0.25 = -0.25$, $x[-2] = -1 + 0.25 + 0.25 = -0.5$, $x[-3] = -0.6875$.

6 Finding $X(z)$ and locating the ROC

HOT 6/45 70 Ch 2+4, 70 Asa 2+4, 75 Ash 3+6, 75 Ch 4+6, 75 Bh 4+6, 73 Ma 1+4

QUESTION AS PRINTED 2075 Chaitra · Q3 · [4+6]

State the properties of ROC. Find the Z-transform and locate the ROC of the signal $x[n] = (0.1)^n u[n] + (0.3)^n u[-n - 1]$.

Method

1. Transform each term with its own ROC.
2. **Mind the sign on the left-sided term:** the standard pair is $-b^n u[-n-1] \leftrightarrow \frac{1}{1-bz^{-1}}$, so a $+b^n u[-n-1]$ in the question picks up a minus.
3. Intersect the two ROCs.

Worked

- $(0.1)^n u[n] \rightarrow \frac{1}{1-0.1z^{-1}}$, ROC $|z| > 0.1$.
- $(0.3)^n u[-n-1] \rightarrow -\frac{1}{1-0.3z^{-1}}$, ROC $|z| < 0.3$.

$$X(z) = \frac{1}{1-0.1z^{-1}} - \frac{1}{1-0.3z^{-1}}, \quad 0.1 < |z| < 0.3$$

- The ring is non-empty because the right-sided pole (0.1) is the *inner* one. It does not contain $|z| = 1$, so this signal has **no DTFT**.

70 Ch and 70 Asa 2+4 : the ROC comes out empty

QUESTION AS PRINTED 2070 Chaitra · Q3 · [2+4]

List out the properties of Region of Convergence. Find the Z-transform and locate the ROC of the signal. $x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{3}\right)^n u[-n-1]$

- First term: $\frac{1}{1+\frac{1}{3}z^{-1}}$, ROC $|z| > \frac{1}{3}$.
- Second term is already in the $-b^n u[-n-1]$ form, so $\frac{1}{1-\frac{1}{3}z^{-1}}$, ROC $|z| < \frac{1}{3}$.
- **Both radii are $\frac{1}{3}$.** The intersection of $|z| > \frac{1}{3}$ and $|z| < \frac{1}{3}$ is **empty**.

The Z-transform of this $x[n]$ does not exist.

- That is a complete, correct answer, and it is the point property 8 of §2.2 is making: an ROC must be a connected, non-empty region.

The siblings

- **75 Ash 3+6** : $x[n] = (0.6)^n u[n] + (0.25)^n u[n]$: **both right-sided**, so both ROCs are “outside” and the intersection is the larger one:

$$X(z) = \frac{1}{1-0.6z^{-1}} + \frac{1}{1-0.25z^{-1}}, \quad |z| > 0.6$$

- **75 Bh 4+6** : $x[n] = (0.25)^n u[n] + (0.6)^n u[-n-1]$: the mirror of 75 Ch:

$$X(z) = \frac{1}{1-0.25z^{-1}} - \frac{1}{1-0.6z^{-1}}, \quad 0.25 < |z| < 0.6$$

- **73 Ma 1+4** : $x[n] = (0.5 + j0.2)^n u[n] + (-j)^n u[-n-1]$.
 - Radii are $|0.5 + j0.2| = \sqrt{0.29} = 0.5385$ and $|-j| = 1$, so **ROC $0.5385 < |z| < 1$** .
 - $z = 0.4$ is **outside the ROC**, so **$X(0.4)$ does not converge**: that is the answer, not a number.
 - $z = 0.7$ is inside:

$$X(0.7) = \frac{1}{1 - \frac{0.5+j0.2}{0.7}} - \frac{1}{1 + \frac{j}{0.7}} = 1.4211 + j2.2198.$$

The intended question verified/added

- This is BB Sir's worked example with one digit changed: his version is $x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{2}\right)^n u[-n-1]$.
- Then the radii are $\frac{1}{3}$ and $\frac{1}{2}$, the right-sided pole is the inner one, and

$$X(z) = \frac{1}{1+\frac{1}{3}z^{-1}} + \frac{1}{1-\frac{1}{2}z^{-1}} = \frac{2z(z - \frac{1}{12})}{(z + \frac{1}{3})(z - \frac{1}{2})}$$

with **ROC $\frac{1}{3} < |z| < \frac{1}{2}$** .

- **In the exam**: give the empty-ROC answer for the signal as printed, in two lines, then solve the $\frac{1}{2}$ version. Both halves earn marks; neither alone is the full answer.

7 Inverse by inspection: $X(z)$ is already a polynomial

PIN 2/45 74 Ash 3+3, 78 Ch 4 · no partial fractions, no ROC decision: 30 seconds if you spot it

74 Ash 3+3

- $X(z) = z^2 \left[1 - \frac{3}{2}z^{-1}\right] (1 + z^{-1})(1 - z^{-1})$.
- $(1 + z^{-1})(1 - z^{-1}) = 1 - z^{-2}$, so

$$X(z) = z^2 \left(1 - \frac{3}{2}z^{-1}\right) (1 - z^{-2})$$

- Expand:
 $z^2 \left(1 - \frac{3}{2}z^{-1} - z^{-2} + \frac{3}{2}z^{-3}\right) = z^2 - 1.5z - 1 + 1.5z^{-1}$.
- Read the coefficients as impulses, **a power of z is an advance**:

$$x[n] = \delta[n+2] - 1.5\delta[n+1] - \delta[n] + 1.5\delta[n-1]$$

- As a sequence: $x[n] = \{1, -1.5, \underline{-1}, 1.5\}$ for
 $n = -2 \dots 1$.

78 Ch 4

- $X(z) = z^2(1 - 1.5z^{-1})(1 - z^{-1})(1 + z^{-2})$.
- $(1 - 1.5z^{-1})(1 - z^{-1}) = 1 - 2.5z^{-1} + 1.5z^{-2}$.
- Times $(1 + z^{-2})$: $1 - 2.5z^{-1} + 2.5z^{-2} - 2.5z^{-3} + 1.5z^{-4}$.
- Times z^2 : $z^2 - 2.5z + 2.5 - 2.5z^{-1} + 1.5z^{-2}$.

$$x[n] = \{1, -2.5, \underline{2.5}, -2.5, 1.5\}, \quad n = -2 \dots 2$$

Why there is no ROC to worry about: a finite-length sequence converges everywhere except possibly $z = 0$ and $z = \infty$ (§2.2, property 4). Both of these have positive and negative powers, so their ROC is $0 < |z| < \infty$.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. $\rightarrow$ reads "leads to / therefore", $+$ reads "and", $=$ reads "is". $\checkmark$ means the property holds, $\times$ that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months:** 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II).
Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 Bh is EX 753 Regular, 74 Ma is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.